

# ANTHONY HO

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## SUMMARY

**Computer Science and Business student at the University of Waterloo (Class of 2026).** Experiences across Software and ML / AI Engineering. Strong computer science fundamentals with proven record of commitment, performance, and impact.

## EDUCATION

**University of Waterloo**, Computer Science and Business Admin. (Double Degree) | GPA: 3.8 / 4.0

Sep 2021 – Aug 2026

**Awards:** 2022 NSERC Undergraduate Student Research Award, 2024 President's Research Award

## SKILLS

**Languages:** Python, Go, PHP, JavaScript, C/C++, SQL

**Tools:** Pandas, PyTorch, PySpark, Docker, Kubernetes, AWS, Polars, Scikit-learn

## EXPERIENCE

### Bree Technologies (YC S21)

*Machine Learning Engineering Intern*

May 2025 – Aug 2025

- Upgraded the core ML pipeline using large-scale data processing optimizations, achieving **>20x** in speed improvements.
- Boosted OCR results accuracy by **>2x** using careful experimentation, prompt engineering, and iteration.
- Re-architected the previous ML model using a stacked ensemble, raising **AUC** to **0.94** and forecasted income by **5M+ / year**.
- Migrated and deployed the new ML infrastructure using FastAPI and Modal, reducing cloud compute costs by **~100x**.
- **Technologies:** *Python, TypeScript, React, SQL, AWS, WandB, Modal*

### TextNow

*Software Engineering Intern*

Jan 2025 – Apr 2025

- Deployed features with no downtime on high-stake, high-traffic endpoints (**5k/s**) using 100% coverage tests and monitoring.
- Resolved a **P0 incident** by debugging improper references after working hours, unblocking teams for the next-business day.
- Jointly architected and rolled out microservices designed for **scale** and **reuse**, including a GeolIP service and a bulk disabler.
- Won the TextNow company hackathon by designing a RAG-based chatbot in Slack, resulting in a \$300 **company award**.
- **Technologies:** *Go, PHP, Kubernetes, SQL, AWS, gRPC*

### Sun Life Financial

*Software Engineering Intern*

Jan 2024 – Apr 2024

- Finalized a SAS-to-Python compiler using **AST traversal**, automating code migration and saving **200+ engineering hours**.
- Used SQL, PySpark, and query optimization to build efficient data pipelines supporting production-scale big data (**TBs**).
- Migrated critical authentication services to AWS using Boto3-based automation, resulting in a **company feature**.
- **Technologies:** *Python, PySpark, SQL, AWS*

### Huawei

*Machine Learning Researcher*

Jan 2023 – Dec 2023

- Researched machine learning methods towards optimizing parameters for the TCP BBR algorithm, contributing to the team's final solution and **paper acceptance into USENIX ATC '24 (< 15% acceptance rate)**.
- Leveraged a variety of ML paradigms such as Causal Forests, HDBScan, XGBoost, and feature selection (PS matching), producing a model with inference error as low as **RMSE = 4%**.
- **Technologies:** *Python, R*

## PROJECTS

*Personal Projects* – see <https://ant52ho.github.io/>

- Projects developed from passion and curiosity. Includes a **blog** (with a backend), a **poop app**, some **hackathon wins**, & more.
- Highlight: **AntRAG** – a self-made Anthony chat available for trusted blog users.
- Bonus: Implemented classical ML algorithms from scratch, including MLPs, Random Forests, hard-margin SVMs, K-Means, logistic regression, EM, and naïve Bayes. Resulted in comparable results to sci-kit learn.